

HW6

1. If the TCP round-trip time, RTT, is currently 30 msec and the following acknowledgements come in after 26, 32, and 24 msec, respectively, what is the new RTT estimate using the Jacobson algorithm? Use $\alpha = 0.9$.

2. Consider the effect of using slow start on a line with a 10-msec round-trip time and no congestion. The receive window is 24 KB and the maximum segment size is 2 KB. How long does it take before the first full window can be sent?

3. In TCP Tahoe suppose that the TCP congestion window is set to 128 KB and a timeout occurs. How big will the window be if the next 10 transmission bursts are all successful? Assume that the maximum segment size is 1 KB.

4. Suppose that the TCP congestion window is set to 18 KB and a timeout occurs. How big will the window be if the next four transmission bursts are all successful? Assume that the maximum segment size is 1 KB.

5. A TCP machine is sending full windows of 65,535 bytes over a 1-Gbps channel that has a 10-msec one-way delay. What is the maximum throughput achievable? What is the line efficiency? (give your answer as xx.x)
 - maximum throughput: _____
 - line efficiency: _____

6. In a network that has a maximum TPDU size of 128 bytes, a maximum TPDU lifetime of 30 sec, and an 8-bit sequence number, what is the maximum data rate per connection?

7. The maximum payload of a TCP segment is 65,495 bytes. Why was such a strange number chosen?

8. If the window size field of the acknowledgement TCP segment is 100 KB, and the congestion window size is 50 KB, how many bytes could the sender transmit next time?

9. What is used at the transport layer to stop a receiving host's buffer from overflowing?

- A. Segmentation
- B. Packets
- C. Acknowledgments
- D. Flow control

10. Which type of service is provided by TCP?

- A. request-reply
- B. acknowledged datagram
- C. reliable message stream
- D. reliable byte stream